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Weeds utilized in the rapeutical purposes by the ethnic tribes of Khunti District(Jharkhand) India

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Abstract : The ethnic-tribes of Khunti district are very expertise in utilizing different weeds as medicine for the treatment of various ailments/diseases. Altogether 22 species of weeds, belonging to 19 families, have been documented during the Ethno-medico-botanical survey in Khunti district. The weeds are used by the local horopaths, vaids, kabirajs, pahans and other knowledgeable indigenous people for the treatment of various ailments/diseases such as jaundice, fever, gastro-intestinal problems, spermatorrhoea, rheumatism and skin problems. Prominent weeds are *Achyranthes aspera, Boerhaavia diffusa, Cynodon dactylon, Euphorbia hirta, Phyllanthus niruri, Commelina benghalensis, Cyperus rotundus, Eclipta alba, Leucas aspera.*

Key Words: Ethno-medicinal weeds, therapeutical, ethnic tribes, horopaths, pahan, expertise, ailments, diseases.

INTRODUCTION

Weeds are the unwanted plants which either grows in the crop or in waste lands on road sides or in remote waste places in forests etc and usually harbor insects, pests and various micro-organisms. Weeds are usually herbaceous plants not valued for use or beauty, growing wild and rank, and regarded as cumbering the ground or hindering the growth of superior vegetation. The term weed is used in a variety of senses, generally centering on a plant that is not desired within a certain context. Weeds are such plants that grow where it is not desired. Weeds are plants out of place and not intentionally sown. Some weeds release into the soil, some poisonous substances that are harmful to crop plants, human beings and animals. A weed decreases the quality as well as yield of agricultural produce. Weeds are very difficult to eliminate because of their certain natural powers. Some weeds have broad leaves while others have narrow leaves. A number of weeds are edible and their parts are used for food and medicines. Khunti district, primarily dominate with different tribal communities, is very rich in its biodiversity where numerous ethno-medicinally important plants grow either in remote forests or in waste lands or even as unwanted weeds in the crop fields. The ethnic tribes that constitute more than 70% of the total population of the district are nature loving and lives either in the forests or in adjoining areas since time immemorial, maintaining their socio-cultural, socio-economic and socio-religious institutions intact. This mass still depends on various forest produces, both major as well as minor, for their subsistence and livelihood. As such this mass is also called as forest dependent population. There is a very close and intricate relationship in between the tribes and the forests. The ethnic tribes of Khunti district are very expertise in utilizing various weeds, either as food or medicine, growing either in waste lands or in the crop fields. They know the nutritional as well as medicinal values of various weeds which are not known to the general urban people. Even such knowledge is still not has been systematically documented either by the Botanists or the Anthropologists. This mass go to the local herbal medicine practitioners (Horopaths, Pahans, Kabirajs, Vaids, Chamains) and other

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knowledgeable persons for the treatment of all sorts of ailments/ diseases. The local herbal medicine practitioners of Khunti district use many weeds as medicine for the treatment of various diseases.

In the present paper attempt has been made to document and enlist the different weeds being used in various therapeutical purposes by the ethnic tribes of Khunti district, Jharkhand.

MATERIALS AND METHODS —

Extensive ethno-medico- botanical field survey in different areas under Khunti district was conducted in different seasons during the years 2009-2011 for the fulfillment of University Grants Commission, New-Delhi, sponsored Minor Research Project entitled "Survey and Inventorization of Medicinal Plants in Khunti district of Jharkhand." Documentation of relevant information and collection of live forms were done following the standard methods(Jain, S.K., 1963[1]; Jain ,S.K. 1964[2]; Jain, S.K. & D.C.Pal;1976[3]: Jain S.K.1995t). During the field survey ethno-medicinally important plants of Khunti district were collected and herbaria have been prepared & preserved in the honey-comb almirah in Ethno-botanical Laboratory, University Department of Botany, Ranchi University, Ranchi. Relevant information and documentation of ethno-medicinal uses of plants were recorded through direct interview with the local horopaths, pahans,vaids, kabirajs and other knowledgeable persons of the area. Collection and documentation of data was done with the help of a questionnaire containing questions on various utilities of plants especially the therapuetical uses, such as local & common name of the plant economic importance, it's medicinal properties, parts used as medicine, diseases for which it is in use, methods of drug preparation & its application, dose, duration etc. Help from local plant collectors and knowledgeable persons were sought in collection of plants and selection of site for survey.

OBSERVATIONS AND RESULTS

In course of the ethno-medicinal field survey in Khunti district under Jharkhand state during the years, 2010-2012, inventorization of the medicinal plants have been done. It has been noticed that the local herbal plant medicine practitioners of Khunti district use many weeds as drug for the treatment of various diseases/ ailments/ disorders. Altogether 22 weed plants have been enlisted belonging to 18 families , that are in use by the local horopaths, pahans, vaids and other knowledgeable persons, as drug for the treatment of various diseases. Such weed plants have been arranged alphabetically giving their botanical name followed by family name in parenthesis, followed with local names/common name in capital letters and then the ethno-medicinal use of the plant.

Abutilon indicum (L) sw.syn. "Malvaceae". KANGHI, Parts used – whole plant, root & bark; Bark paste used in cuts & wounds- it checks bleeding. Root infusion in fever, piles & as nerve tonic, bark used for increased secretion & discharge of urine.

Acalypha indica L. ***A. spicata***, ***A. ciliata***. "Euphorbiaceae". KULPHI, INDIAN ACALYPHA. Parts used – Whole plant. Plant decoction in loosening the intestine, root & leaves used in skin diseases, in gastro-intestinal problems, in ulcers & bronchitis, leaves also used in snake bites & scabies, root is emetic –used in controlling vomiting.

Achyranthes aspera L. "Amaranthaceae". CHIRCHITA, PRICKLY CHAFF- FLOWER TREE. Parts used – Root, leaves, seeds, whole plant. Stem paste mixed with little water inhaled orally. One-two tea-spoon twice daily for 3-4 weeks-it dissolves kidney stone gradually. Fresh leaves chewed in empty stomach- it acts as a good appetizer. Whole plant used in the treatment of leprosy.Extract of whole plant inhaled orally- useful in increasing urine discharge. Root used as tooth brush- it is good for strengthening gums & gives relief in tooth problems. The plant is also used for skin diseases and bleeding piles.

Adiantum phillippense l. "Adiantaceae". KUKUR JARI. Parts used – Rhizome & fond. Paste of the rhizome and fond is applied for dog bite, scorpion bites and centipede bite. Mediation for 5-7 days, twice or thrice daily.

Amaranthus spinosus L. "Amaranthaceae". KATELI CHAULI. Parts used - Root, stem, leaves and whole plant. Two tea-spoon root juice inhaled continuously with warm water before bed time for 30-45 days –it breaks and dissolves gravel & comes out through urine. Two-three tea-spoon of filtered root juice is given with cold water in the morning for a few days to persons suffering from

urinary problems, such as difficulty in passing urine (Dysuria). Infusion of young shoots is used for the treatment of Eczema. Roots of ***A. spinosus*** ground into paste after mixing with Sonth (Dried ***Zingiber officinale***). The paste is given as a medicine for cure of indigestion.

Argemone mexicana Linn. "Papaveraceae", PILA DHATURA; YELLOW MEXICAN POPPY. Parts used— Whole plant, root, leaves, milky juice of fresh plant, seed oil. Roots are used in guinea- worm infestation, skin diseases, leprosy, inflammations, and all types of poisoning, constipation and malarial fever. The leaves are used in cough, wounds, ulcers and skin diseases. The seeds are useful in asthma, cough, skin diseases, leprosy, ulcers, wounds, dental caries, and constipation. The latex is used in skin problems, jaundice, leprosy, blisters, conjunctivitis, inflammations, burning

sensation and malarial fever. The seed oil is useful in wounds, leprosy, skin diseases, constipation and indolent ulcers.

Boerhavia diffusa L. **"** Nyctaginaceae" . KHAPRA SAG, PUNARNAVA. SPREADING HOG-WEED. Parts used – Whole plant. The plant is useful in all types of inflammations, leucorrhoea, scabies, cardiac disorders, jaundice, anemia, constipation, cough, bronchitis and general debility. In menstruation problems — 10 gm. of plant paste in rice water (water obtained after washing rice) with little amount of honey, given twice daily, against excessive profuse discharge during the menses (menorrhagia.). The plant extract is used in dropsy. Root filtrate is used in stomach pain, epilepsy and fever. Dose – 1-2 tea-spoons full given only once everyday for 10-15 days. Plant extract is used for the treatment of Ascites of liver (An abnormal accumulation of fluid in the liver).The plant is useful for the remedy from jaundice.

Cassia tora **Linn.** ***Cassia obtusifolia*** **Linn.** ***C. hirsuta*** **Linn. Fam. "Caesalpiniaceae".** CHAKUNDA, FOETID CASSIA**.** Parts used – Root, leaves & seeds. Tender leaves & tiny shoots used against bowl problems. Seeds used in skin diseases & ringworm, seed paste applied in affected areas. Medication- 3-4 times daily for 10-12 days. Fried leaves applied against itch & fowl ulcers. Dose- 2-3 times daily for 5-7 days. Leaves used in wounds. It

helps in pus formation, thus helps in early cure. Soft musk of leaves in oil is made (poultice) and put on the wounds. It is kept for 2-3 days. It is tied with cloth if necessary. The pus formation in the wounds is activated and released soon. For epilepsy –The whole plant is crushed & the extract is obtained & filtered. The filtered extract is administered for the treatment of Epilepsy. Mediation — 2 Tea spoon plant extract daily once, continuously for 15-20 days

Chenopodium album **L."Chenopodiaceae".** BHATHUA SAG, WILD SPINACH. Parts used – Whole plant. Whole plant protects body from peptic ulcer and intestinal disorders. The plant is effective against indigestion, anthelmintic and laxative.The plant is also useful in seminal weakness, cardiac disorders, helminthiasis and general debility**.**

Commelina benghalensis L. "Commelinaceae" . KANA SAG. Parts used – leaves & roots. Roots as antidote to snake bite. Plant used in leprosy. Leaves used against urinary problems. Medication- 5-7 fresh leaves chewed daily for 30-45 days. It cures urinary problems.

Cynodon dactylon **(L.)** Pers."Poaceae". DUB-GHAS, BAHAMA GRASS. Parts used – whole plant. Few drops of the leaf-juice are dropped inside the nostrils to stop bleeding. Two teaspoon of the leaf-juice is taken with cold

water regularly before bed-time for its cooling properties. Two-three tea-

spoon of the leaf-juice is taken with water for 2-3 times, with an interval of 2-3 hours to cure food poisoning. Root decoction used for the treatment of syphilis. Root decoction is diuretic. Used to increase secretion & discharge of urine. Root decoction also used in dropsy i.e. an excessive accumulation of clear or watery fluid in any of the tissues or cavities of the body. Root infusion used for stopping bleeding from piles.

Cyperus rotundus **L.** "Cyperaceae". NAGAR MOTHA, MOTHA, NUT GRASS. Parts used – tuberous rhizome. In intestinal worms (Anthelmintic) — Rhizome of *Cyperus rotundus* and root of *Cynodon dactylon* are mixed in equal proportion, crushed and extracted the juice. Two three teaspoon of the juice is taken 3-4 times with an interval of 3-4 hours to expel the intestinal worms. Rhizome paste is eaten for the anthelmintic action. About 20gm. of the paste is eaten in the first dose which is followed by

that of 10gm.The course is 3 times a day for 2-3 days. However, the dose is reduced to

half in case of children. Five-eight gm. of the rhizome is consumed in the morning for a week or more to increase the appetite. The rhizome is also useful in wounds, ulcers, fever, vomiting, malarial fever, diarrhoea, dysentery, skin diseases, scabies cough and general debility.

Eclipta alba Hassak. E. Prostrata Linn. " Compositae". BHANGRA. TRAILING ECLIPTA. Parts used – Whole plant. Plant paste is used

against scorpion bite. The plant is used in gastropathy, skin diseases,

wounds, leprosy, jaundice and for strengthening of the gums. Seeds are good for increasing sexual vigor. Plant is also good for blackening and strengthening of the hair.

Euphorbia hirta L., ***E. Pilulifera***. " Euphorbiaceae", DUDHI. Parts used – whole plant, leaves, latex. Whole plant used in cold, asthma, to nursing mothers as a galactogue, in urino-genital tract diseases. Leaf extract mixed with sugar given for dysentery. Dose- 2-3 tea-spoon thrice daily for 5-7 days. Latex in warts & cuts. 1-2 tea-spoon leaf extract with honey orally administered in the morning for a month for leucorrhoea. Latex used in eye problems

Leucas cephalotes (Roth.) Spreng. "Lamiaceae", "Labiatae". GUMA. Parts used – whole plant, leaves & flowers. Fresh leaves and seeds mixed almost equally and made to paste & are applied for ulcerous wounds of cattle. Leaf juice is used for the treatment of headache & vomiting (Migraine). Dose – 1-2 teaspoon twice daily for 2-3 days. 1-2 drops of root juice is put in the nostril. Leaf juice is applied externally for the treatment of psoriasis and chronic skin eruptions. The juice is applied externally on the affected skin as

a lotion 3 –4 times daily continuously for 10-15 days. Flowers with honey are given to children suffering from cough & cold. Leaf or root juice is given in drops for different eye complaints. Fresh plant juice is used in scabies. Plant decoction is useful in malarial fever.

Oxalis corniculata L. "Oxalidaceae". AMTIA SAG, TINPATIA, INDIAN SORREL. Parts used – Whole plant. About 10-15 ml leaf juice raw is given twice a day for 3-4 days against chronic dysentery and diarrhoea. The plant is used as a medicine for gastric problems, headache, for bleeding piles, for indigestion (dyspepsia), as anodyne (kills pains), in deficiency of blood (anaemia), for painful & difficult menstruation (dysmenorrheal), as hemorrhage (to check profuse bleeding from any part of the body). Leaf infusion is used for the treatment of different eye problems. Method –leaf infusion is used as eye drop against itching of eye lids.

Portulaca oleracea L. "Portulacaceae". BARA LANIAYA. Parts used – Fresh leaves, stem & seeds. Leaves are consumed to cure scurvy. Juice of leaves and stem is applied on soles of feet and palms to relieve burning sensation. Seed paste is applied over boils and ulcers to prevent microbial growth. A cotton plug wetted in leaf juice is placed on tooth to relieve toothache. Leaves are eaten in liver diseases.

Phyllanthus niruri Linn. "Euphorbiaceae". Hazardana. Parts used- roots, leaves and whole plant. Plant is used against different problems in genito- urinary tract. Leaves decoction is used in dysentery. Fresh root used for jaundice.

Scoparia dulcis L."Scrophulariaceae". CHINISAKAM, SWEET BROOM. Parts used – Whole plant, leaves & roots. The plant is used in diabetes, blood pressure, wounds, burns and headache. Tea of the plant is taken to reduce swellings & pains. Plant decoction is administered orally for three days each month during menstruation. It acts as a contraceptive and /or to induce abortion. Root or leaf extract mildly heated and administered for dysentery. Leaf decoction or hot water infusion is taken orally for stomach pain, for menstrual disorders, as an aid in child birth, as a blood purifier, in heart problems, in liver & stomach disorders, and in sexually transmitted diseases. The plant paste is applied on the forehead for the remedy of headache. The aerial part of the plant is crushed and the aqueous extract is used for spermatorrhoea. Dose – 5-10 ml extract is administered orally twice a day for 5-7 days against spermatorrhoea.

Sida cordifolia L. "Malvaceae" . BALA, BARIALA, FLANNEL WEED, HEART LEAF SIDA. Parts used – Whole plant. Plant is used for the treatment of inflammation of the oral mucosa, asthmatic bronchitis, nasal congestion and other respiratory problems. Plant decoction is given during Asthmatic attack. Root infusion is given in urinary

troubles. Powdered root is used with milk in Leucorrhoea. Leaves are used in diarrhoea and dysentery.

Solanum indicum L.S. nigrum Linn. Solanum surrattense Buem. F., ***S. Xanthocarpum*** Schrad. & Wendl. , ***S. anguivi*** Linn. "Solanaceae", MAKOI, RANGAINI, KANTAKARI, BLACK NIGHT SHADE. Parts used – whole plant. Fruits felicitate seminal ejaculation; alleviate worms, itching and fever & reduce fat. Dried fruits are smoked in the form of cigarette and the smoke is held up in the mouth cavity for sometime, it ameliorates the dental infection. The fumigation of Kantakari is helpful in piles. Plant/fruit/roots paste is applied in swollen and painful joints in arthritis, it reduces the pain & swellings effectively. Root decoction works well in cough when given with honey. The plant is extremely beneficial in the treatment of cardiac diseases associated with edema. Root mixed with few grains of black Pepper is applied externally for chest pain. Root *mixed* with that of *Alternanthera sessilis* (Garundi arak), in equal amounts in water & given in urinary problems. Leaf juice with Black pepper is used in rheumatic problems.

Xanthium strumarium L. " Asteraceae". SINGHI. Parts used- whole plant, leaves & fruits. Leaf juice is put into the ear to relieve ear-ache. Leaves boiled in water or warmed as such are bandaged on abdomen to promote urination. Fruit is attached to ear lobe as ear pendant to relieve headache. Ash of whole plant or leaves is boiled with Coconut or Neem oil and the medicated oil is applied on ulcers and wounds for fast healing.

DISCUSSION —

Plants are being used as medicine by the human being from pre-historic period and are used for health care and treatment of various diseases, disorders throughout the world (Jain, S.K. & D.C.Pal; 1976[3]:Jain S.K.1995t : Harshberger,J.W.1896u JainS.K.&R.R.Rao1983v). Treatment of spermatorrhoea,various gynecological problems, gastro-intestinal disorders by ethno-medicinal plants have been worked out. (Rameshwar Mahto & H.B.Sahu 2007w : Sahu, H.B., S.L.Bondya, Jyoti Kumar & H.P.Sharma 2003x : Rama Rao, N. & A.N.Henry 1996y). A comprehensive list of ethno medicinal plants of Eastern Ghats in Andra Pradesh has also been published(Rama Rao, N. & A.N.Henry 1996[1]R"). Treatment of Menorrhagia prevailing among the indigenous women of Chandil block of Saraikela Kharsawan district of Jharkhand by horopathy has been reported (Rameshwar Mahto & H.B.Sahu 2008[11]):.

The present paper is mainly confined with the collection & documentation of information about different weeds found in Khunti district being used by the local herbal medicine practitioners for the treatment of various diseases as mentioned earlier. Scientific identification and preservation of such weeds have been done following the standard methods ***(Jain, S.K.,*** (1963[1]; Jain,S.K. 1964[2]; Jain,S.K. &Pal, D.C. 1976[3]; Jain,S.K. 1995t).The pharmaceutical properties of these weeds are still unnoticed, unexplored and not commonly known, as a result, the common people in general and the urban populace in particular are still not in use of such weeds for various pharmaceutical purposes. The information recorded here needs proper documentation and wide publicity so that these could be used by the deprived mass and to develop ways and means for protection and cultivation of these so called weeds or unwanted plants which are neglected only due to their harmful activities. This will definitely be very useful for the mankind as these plants are easily available, cheap & having no side effects.

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